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Tectonic implications of new Late Cretaceous paleomagnetic constraints from Eastern Liaoning Peninsula, NE China

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Abstract

[1] A paleomagnetic study has been carried out in the east of the Tan-Lu fault, in Liaoning Province, NE China, to understand the timing of Tan-Lu fault activity. Samples aging from Early Paleozoic to Late Mesozoic from 51 sites have been analyzed. Paleozoic and Late Permian-Early Triassic rocks are remagnetized by the recent geomagnetic field; however, Late Cretaceous (between 118 and 83 Ma) red tuffaceous sandstone passes a positive fold test, shows no Present Earth Field characteristic remanent magnetization carried by both magnetite and hematite, presents a solo normal polarity, and thus provides the only reliable paleomagnetic data in this study. The paleomagnetic pole calculated from these rocks (λ_p = 59.4°N, ϕ_p = 205.5°E, and A_{95} = 7.3°) is statistically undistinguishable from all available Cretaceous paleomagnetic data from Eastern Liaoning-Korean Peninsula, indicating

that these areas may belong to a single tectonic unit, here named the East Liaoning-Korea (ELK) Block, at least since the Late Cretaceous. Conversely, a significant discrepancy between the ELK Block and Chinese Block (i.e., North and South China Blocks) characterized by a differential rotation (22.5° ± 10.2°) with a negligible latitudinal displacement (0.8° ± 6.1°) is demonstrated. This result indicates, first, that the left-lateral displacement along the Tan-Lu fault, if any, must have occurred before the Late Cretaceous and, second, that the Korean Block can not be considered as a rigid part of North China Block. Sedimentological and structural evidence show that the Meso-Cenozoic triangle shaped plain, consisting of Songliao, Xialiaohe, Sanjiang, Zeya, and other smaller basins developing in northeast China and southeast Russia, south of the Mongol-Okhotsk Belt, experienced a heterogeneous rifting from the Late Jurassic to Tertiary. The variable amount of extension, larger in the northeast (~300 km) than in the southwest (~80 km), is probably related to the clockwise rotation of the ELK Block centered at the south of the Bohai Bay Basin. This result shows that a significant segment of the East Eurasian margin (more than 1000 km long) experienced differential rotation in Cenozoic times.

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